

ENERGY STAR Portfolio Manager Integration

PREPARED BY Energy Hippo implementation team

This document explains how the synthetic customer maps to ENERGY STAR Portfolio Manager (PM) so that a future integration run is mostly configuration, not modeling. The synthetic data is intentionally shaped to match the PM model that the EEMSuite EnergyStar integration already generates and consumes.

Where this is grounded

- **EEMSuite integration model** (the XSD-generated C# the integration uses):
`energyhippo/EEMSuite/src/Integrations/EEMSuite.Integrations.EnergyStar/Models/Generated — PropertyType, PrimaryFunctionType, GrossFloorAreaType, OccupancyType, and the use-detail value types.`
- **EEMSuite mapping code:**
`energyhippo/EEMSuite/src/Tasks/EEMSuite.Tasks.EnergyStar/Mapping/PropertyMapping.cs` — maps an EEM facility row to a PM PropertyType.
- **ENERGY STAR public reference:** "U.S. Property Types, Definitions, and Use Details" (Oct 2024), and the PM web services property API. Links in [Sources](#).

The PM property model

A PM **Property** is one benchmarked building or campus. It carries:

PM field	Source	Notes
name	building/site display name	1–80 chars
primaryFunction	PrimaryFunctionType enum	Must be an exact enum value or it cannot be parsed
grossFloorArea	GrossFloorAreaType	value + units (Square Feet / Square Meters)
yearBuilt	int	
occupancyPercentage	OccupancyType	Discrete 5% increments (0, 5, ..., 100)
isInstitutionalProperty	bool	True for campus buildings
address	AddressType	

A Property then has one or more **Property Uses** (e.g. Office, Library, Multifamily Housing). Each Property Use carries **Use Details** — the characteristics PM uses for metrics. A single building can hold several uses (e.g. an office building with a classroom wing).

Use-detail wrapper

Every use detail derives from `UseAttributeBase`, so each value can carry:

- `value` (the measurement), and `units` where applicable.
- `currentAsOf` (date the value became current).
- `temporary` (flag for provisional values).
- `default` (read-only; PM marks values it defaulted).

Value carriers in the generated model: `UseIntegerType` (`int ≥ 0`), `UseDecimalType` (`decimal ≥ 0`), `UseStringType`, `UseYesNoType` (Yes/No), `GrossFloorAreaType`, plus typed enums such as `ResidentPopulationType`, `OnsiteLaundryType`, and `AnnualResearchExpenditureType`.

Key enum/unit values used by this dataset:

- `GrossFloorAreaUnitsType`: Square Feet, Square Meters.
- `ResidentPopulationType`: No specific resident population, Dedicated Student, Dedicated Military, Dedicated Senior/Independent Living, Dedicated Special Accessibility Needs, Other dedicated housing.
- `OccupancyType`: 0, 5, 10, ... 100. `PercentHeated / PercentCooled`: 0, 10, ... 100.
- `PoolSizeTypeValue`: Recreational (20 yards x 15 yards), Short Course (25 yards x 20 yards), Olympic (50 meters x 25 meters). `PoolTypeValue`: Indoor, Outdoor. The Recreation and Aquatics Center pool is a short-course indoor pool.

Three grouping models in this dataset

PM supports more than one way to group buildings; the synthetic customer exercises three deliberately:

1. **Campus** — Northlake Main Campus is a College/University **campus parent** of 545,000 sq ft. Each campus building (Admin Hall, Science Center, Library, Student Center, Lakeview Residence Hall, Recreation and Aquatics Center, Parking Structure) is its own **child property** with its own `primaryFunction` and `uses`, rolling up to the campus. The Parking Structure is benchmarked as its own child property and is excluded from the campus floor area.
2. **Multifamily property** — Cedar Row Apartments is one PM-benchmarked Multifamily Housing **property** (124 units, `Dedicated Student`); Cedar Row A, Cedar Row B and Cedar Row Common House are its physical **buildings within the property**.
3. **Standalone property** — the Central Plant is its own `Other - Utility` property, benchmarked for the Northlake Thermal Plant company and excluded from the campus boundary.

Campus Grounds and Cedar Row Carport and Grounds are outdoor service areas and have no PM property.

This is captured in `data/scenarios/demo-university-v1.yaml` under `energyStarProperties` (13 entries), keyed by `siteId / buildingId` with a `role` (`campus_parent`, `child_property`, `multifamily_property`, `building_within_property`, `standalone_property`).

Synthetic building → PM mapping

Building		Property uses
Northlake Main Campus (site)	College/University	College/University, 545,000 sq ft with parking excluded (Enrollment, FTE Workers, Grant Dollars)
Admin Hall	Office	Office + Other - Education (classroom wing)
Science Center	Laboratory	Laboratory + Other - Education
Library	Library	Library
Student Center	Food Service	Food Service + Other - Entertainment/Public Assembly
Lakeview Residence Hall	Residence Hall/Dormitory	Residence Hall/Dormitory (150 rooms)
Recreation and Aquatics Center	Fitness Center/Health Club/Gym	Fitness Center/Health Club/Gym + Swimming Pool
Parking Structure	Parking	Parking
Cedar Row Apartments (site)	Multifamily Housing	Multifamily Housing (124 units, Dedicated Student) + Other - Recreation (Common House)
Cedar Row A / B	Multifamily Housing	Multifamily Housing (per-building)
Cedar Row Common House	Other - Recreation	Other - Recreation
Central Plant	Other - Utility	Other - Utility (standalone property outside the campus boundary)

Every primaryFunction and function string above is a verbatim PrimaryFunctionType enum value, so ResolvePrimaryFunction parses it instead of falling back to Other.

Use details by property type

Required vs optional per the ENERGY STAR reference (Oct 2024). Optional details do not affect metrics but are still accepted. For Fitness Center/Health Club/Gym, Swimming Pool, Parking, Other - Recreation and Other - Utility the required vs optional split has not been checked against the reference; those rows list only the use details the scenario supplies, by PM name where the manifest key table below defines one and by manifest key otherwise.

Scored is a property-type rule used throughout the packet: in the US a 1–100 ENERGY STAR score exists for Office, Residence Hall/Dormitory and Multifamily Housing. Library is score-eligible only in Canada; every other type in this dataset is metrics only. Per energystar.gov, a property can receive a score only when more than 50 percent of its gross floor area (excluding parking) is an eligible type and the combined floor area of uses without a score does not exceed 25 percent. Lakeview Residence Hall and Cedar Row Apartments are the properties expected to receive a score. Admin Hall exercises the second test: it is Office-primary, but its 25,000 sq ft Other - Education classroom wing is 29 percent of its floor area, so it gets metrics only.

Property type	Scored	Required use details	Optional use details
College/University	No	Gross Floor Area	Weekly Operating Hours, Enrollment, Number of FTE Workers, Number of Computers, Grant Dollars
Office	Yes	Gross Floor Area ($\geq 1,000$), Weekly Operating Hours (≥ 30), Number of Workers on Main Shift (≥ 1), Number of Computers (≥ 1), Percent That Can Be Cooled, Occupancy ($\geq 55\%$ to certify)	Percent That Can Be Heated
Laboratory	No	Gross Floor Area	Weekly Operating Hours, Number of Workers on Main Shift, Number of Computers
Library	No (Canada only)	Gross Floor Area	Weekly Operating Hours, Number of Workers on Main Shift, Number of Computers, Percent That Can Be Heated, Percent That Can Be Cooled
Food Service	No	Gross Floor Area	Weekly Operating Hours, Number of Workers on Main Shift, Number of Computers
Multifamily Housing	Yes	Gross Floor Area, Total Residential Living Units (≥ 20), Low-/Mid-/High-rise unit counts, Number of Bedrooms, Occupancy ($\geq 80\%$ to certify)	Resident Population Type, Government Subsidized Housing, Laundry Hookups (in-unit / common), Percent Heated, Percent Cooled
Residence Hall/Dormitory	Yes	Gross Floor Area ($\geq 5,000$), Number of Rooms (≥ 5), Percent Heated, Percent Cooled	Computer Lab, Dining Hall
Fitness Center/Health Club/Gym	No	Not checked. Scenario supplies Weekly Operating Hours, Number of Workers on Main Shift, Number of Computers, Percent That Can Be Cooled, Percent That Can Be Heated	Not checked
Swimming Pool	No	Not checked. Scenario supplies <code>sizeOfPool</code> , <code>monthsInUse</code> , <code>indoorOrOutdoorPool</code>	Not checked
Parking	No	Not checked. Scenario supplies <code>completelyEnclosedParkingGarageFootage</code> , <code>partiallyEnclosedParkingGarageFootage</code> , <code>openParkingLotFootage</code> , <code>supplementalHeating</code>	Not checked
Other - Education	No	Gross Floor Area	Weekly Operating Hours, Number of Workers on Main Shift, Number of Computers
Other - Entertainment/Public Assembly	No	Gross Floor Area	Weekly Operating Hours, Number of Workers on Main Shift, Number of Computers
Other - Recreation	No	Not checked. Scenario supplies Weekly Operating Hours, Number of Workers on Main Shift	Not checked

Property type	Scored	Required use details	Optional use details
Other - Utility	No	Not checked. Scenario supplies Weekly Operating Hours, Number of Workers on Main Shift	Not checked

Residence Hall/Dormitory is assigned to Lakeview Residence Hall (150 rooms, a computer lab and no dining hall), which exercises the Number of Rooms / Computer Lab / Dining Hall details. Cedar Row stays Multifamily Housing with Dedicated Student population because it is an apartment complex, not a dorm.

Manifest field names → PM use-detail names

The useDetails keys in the manifest map to PM use details as follows:

Manifest key	PM use detail
weeklyOperatingHours	Weekly Operating Hours
numberOfWorkersOnMainShift	Number of Workers on Main Shift
numberOfComputers	Number of Computers
percentThatCanBeCooled / percentThatCanBeHeated	Percent That Can Be Cooled / Heated
enrollment	Enrollment
numberOfFTEWorkers	Number of Full Time Equivalent (FTE) Workers
grantDollars	Grant Dollars
totalNumberOfResidentialLivingUnits	Total Number of Residential Living Units
residentialLivingUnitsLowRise / MidRise / HighRise	Residential Living Units in a Low-/Mid-/High-rise Setting
numberOfBedrooms	Number of Bedrooms
residentPopulationType	Resident Population Type
governmentSubsidizedHousing	Government Subsidized Housing
numberOfLaundryHookupsInAllUnits / InCommonAreas	Number of Laundry Hookups in All Units / Common Area(s)
numberOfRooms	Number of Rooms
computerLab / diningHall	Computer Lab / Dining Hall

The Swimming Pool keys (sizeOfPool, monthsInUse, indoorOrOutdoorPool) and the Parking keys (completelyEnclosedParkingGarageFootage, partiallyEnclosedParkingGarageFootage, openParkingLotFootage, supplementalHeating) are not yet mapped to PM use-detail names here.

Integration hooks (already present in EEMSuite)

- PropertyMapping.ResolvePrimaryFunction(string) parses the function string by its XML enum value and falls back to PrimaryFunctionType.Other (a safe, unscored default) on no match — so an exact string matters.
- PropertyMapping.ResolveOccupancy(int) clamps 0–100 and rounds to PM's 5% scale. The synthetic occupancyPercentage values are already multiples of 5.

- Building mapping sets `grossFloorArea.units` by country (US → Square Feet).
- Property uses are serialized via `PropertyUseRequest` + `PropertyUseXmlBuilder` and read back via `PropertyUseDocument` / `PropertyUseDetailEntry`.

Not yet covered (follow-on work)

- **Meter association + consumption upload:** pushing monthly usage to PM uses `meter.xsd`, `association.xsd`, and `meterConsumption`. The synthetic meters and bills already exist (see the manifest accounts and provider configs); the PM side still needs meter → property association and consumption payloads.
- **Metrics read-back:** validating PM-returned metrics (Site/Source EUI, ENERGY STAR score) against expected synthetic values.
- A **negative scenario:** a building whose `primaryFunction` is an unknown string, to prove the `Other` fallback path.

Sources

- [List of Portfolio Manager Property Types, Definitions, and Use Details — ENERGY STAR](#)
- [U.S. Property Types, Definitions, and Use Details \(Oct 2024, PDF\)](#)
- [Portfolio Manager web services — Property API](#)